CHEM-SHIELD

An AI-Assisted Cyber-Physical Early-Warning and Threat Detection Framework for Simulated Defense-Industrial Operational Technology

October 2026

Executive Summary

CHEM-SHIELD is a defensive cybersecurity research project that studies whether network evidence and physical-process evidence, once correlated, give an operator an earlier and more useful warning of suspicious activity in an Operational Technology (OT) environment. Everything runs inside an isolated virtual laboratory on the loopback interface. No real military, industrial, SCADA, or PLC equipment is accessed, scanned, or modified.

The project has so far produced a working end-to-end prototype made of six parts and a shared publisher:

- **Part 1, virtual OT laboratory:** a Modbus TCP virtual PLC, a physical-process simulator, and an operator HMI with test-event controls that create labelled abnormal conditions.
- **Part 2, network security:** adapters for Zeek, Suricata, and Modbus-monitor records, plus a network anomaly model trained on a normal baseline.
- **Part 3, physical machine learning:** an Isolation Forest trained on normal telemetry, with scripted evaluation on synthetic scenarios.
- **Part 4, correlation and risk:** same-asset, 10-second correlation and a transparent weighted risk score with a persistent incident register.
- **Part 5, OT monitor:** sensor validation and engineering rules combined with the Part 3 model.
- **Part 6 and Admin console:** an operator-confirmed simulated response and a role-based supervisory console with an audit log.

The first measured results are mixed and are reported as they are. The physical model detects abrupt and unstable behaviour well (about 90 to 97 percent per scenario) but misses slow drift and stealthy in-range changes, and its false-positive rate is too high on the current data. These findings are useful: they show exactly where network evidence adds value and they define the planned enhancements described in Section 11.

1. Introduction

1.1 Background: from information systems to cyber-physical systems

Most people think of cybersecurity as the protection of data: passwords, files, databases, and websites. In that world, the worst outcome of an attack is usually lost, stolen, or corrupted information. But many important systems do not only process information. They also move, heat, pressurise, cut, and mix things in the physical world. A factory production line, a water-treatment plant, a power station, a chemical process, and an ammunition or equipment manufacturing line are all examples. The computers that run such equipment are called **Operational Technology (OT)**.

OT is the combination of hardware and software that monitors and controls physical devices and processes. Its main parts are easy to describe:

- **Sensors** measure the physical world (speed, temperature, pressure, level, vibration).
- **A Programmable Logic Controller (PLC)** reads the sensors, applies control logic, and sends commands to the equipment, repeating this many times per second.
- **Actuators and machines** (motors, valves, heaters, pumps) carry out those commands.
- **An HMI or SCADA system** shows the operator what is happening and lets the operator change settings such as a target speed.

These parts talk to each other over industrial protocols. One of the most common is **Modbus TCP**, where a controller exposes numbered registers and any connected device can read or write them. Such protocols were designed decades ago for reliability and simplicity, in closed networks. They usually have no built-in login and no encryption. A device that can reach the PLC and speaks the protocol correctly is, by default, treated as legitimate.

When systems of this kind are connected to modern networks, a cyber event can become a **physical event**. A changed setting does not only alter a record; it can make a real motor overspeed, a temperature rise, or a pressure spike. This combination of the digital and the physical is called a **cyber-physical system**, and it is the setting of this research.

1.2 Why OT security is different from IT security

Security methods built for office networks cannot simply be copied into industrial settings, because the priorities are different.

Aspect	Information Technology (IT)	Operational Technology (OT)
Main priority	Confidentiality of data	Safety and availability of the process
Worst outcome	Data loss or theft	Equipment damage, downtime, or danger to people
Equipment life	About 3 to 5 years	Often 15 to 30 years

Aspect	Information Technology (IT)	Operational Technology (OT)
Patching	Frequent and routine	Rare, because stopping the process is costly or risky
Protocols	Modern, usually authenticated and encrypted	Often legacy, such as Modbus, with little or no authentication
What must be watched	Network traffic, hosts, users	Network traffic and the behaviour of the physical process

The last row is the key point of this project. In OT, watching the network is necessary but not sufficient, because the thing that finally matters is what the machine actually does.

1.3 The limitation of watching only one source

Two kinds of monitoring are commonly used, and each one answers a different question.

Network monitoring (for example Zeek or Suricata) answers: "who sent what to whom, and when?" It can detect an unusual source, an unexpected function code, or a burst of write commands. Its weakness is that a command can be **perfectly valid at protocol level** and still be harmful. A write of a new target speed to the PLC looks identical whether it came from the real operator or from an unauthorised sender. The network layer sees the message but not its physical consequence.

Physical-process monitoring (for example a machine-learning model on telemetry) answers: "is the machine behaving normally?" It can notice that speed, temperature, or pressure has moved away from the learned normal pattern. Its weakness is that an unusual reading has many possible causes: wear, a failing sensor, ambient conditions, a maintenance task, or a legitimate operator change. Without network context the monitor cannot say whether a communication event is involved, and without any way to rank alerts it may raise many that are not important.

Neither source alone answers the question an operator really needs answered: **was there a suspicious communication event, and did it produce an unusual physical consequence on the same asset, at about the same time?** Answering that question requires combining the two kinds of evidence.

1.4 The idea behind CHEM-SHIELD

CHEM-SHIELD is a small, controlled framework that tests exactly this combination. In plain words:

- A **simulated machine and PLC** run in a virtual laboratory and produce both network traffic and physical telemetry.
- A **network layer** turns suspicious communication into a structured network event with a score.

- A **machine-learning layer** learns what normal physical behaviour looks like and turns deviations into a structured physical event with a score.
- A **correlation engine** links the two events when they concern the same asset within a short time window.
- A **risk engine** turns the linked evidence into one transparent score and a level, and the **dashboard** shows it to the operator as an early warning.

A worked example, using the default settings of the current prototype, shows why this helps. Suppose that at 09:30:00 an unusual write to the target-speed register of PLC-01 is seen, and the network model gives it a score of 0.9. At 09:30:04 the telemetry shows speed and temperature moving away from normal, and the physical model also gives a score of 0.9. The risk engine computes 0.40 x network score + 0.60 x physical score:

Evidence available	Calculation	Score	Level
Network event only	0.40 x 0.9 + 0.60 x 0	0.36	MEDIUM
Physical event only	0.40 x 0 + 0.60 x 0.9	0.54	MEDIUM
Both, correlated (same asset, within 10 s)	0.40 x 0.9 + 0.60 x 0.9	0.90	CRITICAL

Table 1. Illustrative scores only; they are inputs chosen for explanation, not measured results.

Each source alone leads to a modest MEDIUM level that is easy to ignore, while the correlated pair of events reaches the highest level. This is the behaviour the research sets out to test, and it is also why the project does not treat machine learning as a decision-maker: an anomaly score means only that something is unusual. Judgement comes from combining independent evidence in a rule-based and fully visible way.

1.5 Context and motivation

Ethiopia has publicly documented defense-industrial and engineering capabilities, and its wider technology sector includes automation, communications, artificial intelligence, and cybersecurity. Industrial and manufacturing environments in any country increasingly rely on networked control systems, so building local skills in OT defence is a reasonable academic goal. The project does not claim any knowledge of the real networks, PLC vendors, or procedures of any specific facility, and it does not use a real facility as a model. It works with a generic simulated motor and process, which is enough because the research question concerns detection and evidence correlation, not the details of a particular production line.

A practical motivation also exists. Real OT equipment is expensive, risky to experiment on, and usually off-limits. A reproducible software laboratory allows students and researchers to study detection ideas safely, repeat the experiments, and share the results.

1.6 Problem statement

Network-security monitoring can detect suspicious communication, but it cannot tell whether a protocol-valid command produced an abnormal physical state. Physical anomaly detection can detect unusual behaviour, but an unusual reading alone has many possible explanations and gives no security context. The research problem is therefore to determine whether combining the two evidence streams improves the identification and prioritisation of suspicious cyber-physical events in a controlled OT simulation.

1.7 Aim and significance

The aim is to design, build, and experimentally evaluate a lightweight, AI-assisted early-warning pipeline that detects suspicious network activity and abnormal process behaviour, correlates them, assigns a transparent risk level, and presents the result on a dashboard. The significance is practical and academic:

- It offers a compact, reproducible OT cybersecurity laboratory suitable for teaching and research without real infrastructure.
- It tests, with measurements, whether evidence fusion gives a more informative signal than isolated monitoring.
- It keeps machine learning explainable by using it as one piece of evidence inside a transparent correlation and risk method.
- It reports limitations honestly, which makes the results a reliable base for later extension.

1.8 Scope and boundaries

The project is strictly defensive. All experiments take place in an isolated virtual laboratory bound to the loopback interface, using only synthetic machines, traffic, and telemetry. No real network is scanned or accessed, no malware or exploit tooling is developed, and abnormal scenarios exist only to evaluate detection. The system gives warnings; it does not control equipment automatically. The single response feature is a simulated stop that a human operator must explicitly confirm (Section 7.6).

1.9 Structure of this document

Section 2 states the research gap and contributions. Sections 3 and 4 give the objectives and research questions. Section 5 describes the architecture, Section 6 the method, and Section 7 the work completed so far. Section 8 reports the first results and Section 9 explains the main design decisions. Sections 10 to 15 present the evaluation plan, the planned enhancements, the work plan, the safety and limitation statements, the deliverables, and the conclusion. The appendices list the repository layout and the demonstration procedure.

2. Research Gap and Contribution

The project does not claim to invent intrusion detection, anomaly detection, machine learning, SIEM concepts, or risk scoring. These are established fields. The contribution is a small, experimentally testable framework that joins them in one controlled OT simulation and measures the value of fusing the evidence. The research gap is:

> *Can a lightweight cyber-physical detection pipeline that combines simulated OT network events with machine-learning process anomalies provide more informative and timely early warnings than either evidence source considered independently?*

Expected contributions:

- A reproducible virtual OT environment for defensive experiments (delivered).
- A normal-process baseline and lightweight ML anomaly detector (delivered; improvement planned).
- A network-event layer producing structured evidence (delivered on replayed data; live capture planned).
- A transparent rule-based correlation method and documented risk model (delivered; sensitivity analysis planned).
- A unified dashboard and supervisory console (delivered).
- Experimental results comparing network-only, physical-only, and fused detection (partly delivered; the fused comparison is the main remaining experiment).

3. Research Objectives

3.1 General objective

To design and experimentally evaluate a lightweight AI-assisted cyber-physical early-warning framework that detects suspicious network activity and abnormal process behaviour, correlates the evidence, assigns a risk level, and presents it through a unified dashboard in an isolated virtual laboratory.

3.2 Specific objectives and their status

No.	Objective	Status
1	Develop a simulated OT environment: virtual PLC, process model, sensors, simple HMI	Done (Part 1)
2	Generate and monitor simulated OT network traffic and identify suspicious events	Partly done: replayed fixtures and adapters; live capture not yet
3	Collect normal telemetry and establish a baseline	Done for synthetic data; baseline from live Part 1 telemetry planned

No.	Objective	Status
4	Implement a lightweight ML anomaly detector (Isolation Forest)	Done; accuracy needs improvement
5	Define a normalised event format shared by all components	Done (Part 4 event contract)
6	Implement rule-based temporal and asset-aware correlation	Done (same asset, 10 s)
7	Implement a transparent risk-scoring mechanism	Done; correlation-strength term and sensitivity study planned
8	Develop a web dashboard for near-real-time monitoring	Done (HMI and Admin console)
9	Evaluate with precision, recall, F1, false-positive rate, latency	Done for the physical model; fused evaluation remaining

4. System Architecture (as Implemented)

The architecture is a five-part chain: simulation, network security, physical machine learning, correlation and risk, and dashboard. Implementation added supporting elements: a shared publisher with an ingest token, a persistent incident register, a role-based Admin console, and an operator-confirmed response.

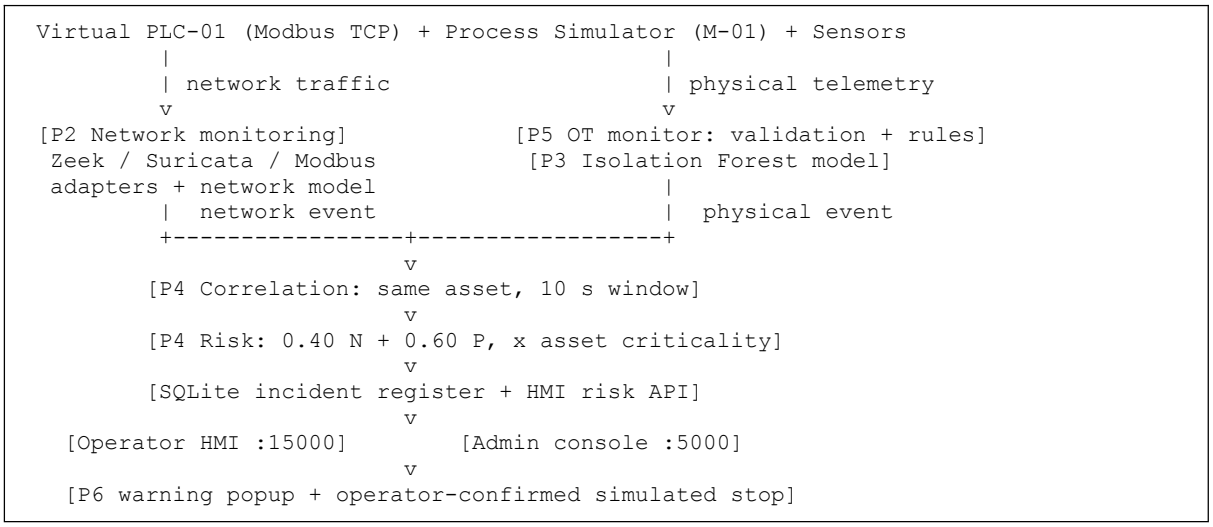


Figure 1. Implemented CHEM-SHIELD pipeline. All services are bound to 127.0.0.1.

All components exchange events in one normalised format. Each event carries a timestamp, an asset identifier (PLC-01), an event type of exactly "network" or "physical", an anomaly score in the range 0 to 1, and an evidence object holding source-specific details. Duplicate event identifiers are suppressed, and publishing to the HMI requires an ingest token and is restricted to loopback addresses.

5. Methodology

5.1 Simulated OT process

The virtual PLC is a Modbus TCP server with an eight-register holding map: motor command, target RPM, actual RPM, temperature, pressure, power, event code, and event value. The HMI writes the command and target, and the separate process simulator reads them, models acceleration, temperature, pressure, and power, and writes the results back to the PLC and to SQLite. A TEST MODE can inject four labelled abnormal events (RPM deviation, temperature increase, pressure spike, sensor inconsistency). Because these events carry ground-truth labels, they allow honest evaluation later.

5.2 Network evidence

Part 2 normalises Zeek connection logs, Suricata EVE records, passive Modbus-monitor output, and exported operating-system security logs into one schema. A network anomaly model, an isolation-forest-style detector trained on a normal network baseline, assigns each record a bounded score. Records are mapped to the simulated asset PLC-01. In the current version this layer replays sanitised fixtures; it does not capture live packets.

5.3 Physical machine learning

Part 3 trains scikit-learn's Isolation Forest only on normal telemetry (2,000 samples, fixed random state 42, contamination 0.05). Isolation Forest was chosen because it needs no labelled attacks, is light enough for a student laboratory, and is well documented. The model uses six features: motor speed, temperature, vibration, and the per-observation rate of change of each. It returns a binary decision and a separate normalised score between 0 and 1. The score is an anomaly indicator, not a probability, and it never declares an attack.

5.4 Correlation and risk

Part 4 validates events and groups them by asset when adjacent timestamps are no more than 10 seconds apart. The risk score is computed as:

```
base_risk  = 0.40 x network_score + 0.60 x physical_score
final_risk = base_risk x asset_criticality      (default 1.0)


final_risk < 0.30           LOW
0.30 <= final_risk < 0.60   MEDIUM
0.60 <= final_risk < 0.80   HIGH
final_risk >= 0.80          CRITICAL
```

When several events join an incident, the maximum score per modality is used, and a later higher score can escalate the incident. An event with no score leaves the incident UNASSESSED rather than inventing a zero. Acknowledging an incident is recorded

separately and never lowers its risk. The weights and thresholds are experimental design choices, not industrial or safety standards.

5.5 Alerts and operator interaction

The operator HMI shows machine state and raises a warning popup when sensor limits are exceeded or a MEDIUM, HIGH, or CRITICAL incident appears. Warnings are informational. Nothing stops or changes the machine automatically.

6. Work Completed So Far

This section records what exists in the project repository. The tables summarise each part; the sub-sections give the detail.

Part	Component	What was built	Status
P1	Virtual OT lab	Modbus TCP virtual PLC, process simulator, SQLite telemetry, Flask operator HMI, test-event controls	Working
P2	Network security	Zeek, Suricata, Modbus-monitor and OS-log adapters; network anomaly model; replay and publish tool	Working on fixtures
P3	Physical ML	Isolation Forest, feature engineering, training, evaluation, and plotting scripts	Working; needs tuning
P4	Correlation and risk	Validation, correlation, risk engine, incident store, risk API	Working
P5	OT monitor	Sensor validation, rule engine, model scoring, publishing	Working
P6	Response	Warning popup; operator-confirmed simulated stop for CRITICAL incidents only	Working
Admin	Supervisory console	Login, three roles, incident review, audit log, live risk view	Working
Common	Publisher and scripts	Loopback publisher with ingest token; one-command launcher; staged chain runner; demo runner	Working

6.1 Part 1: virtual OT laboratory

- Standalone Modbus TCP PLC server (pymodbus) with a documented register map; an OpenPLC runtime can replace it later without changing the other parts.
- Process simulator modelling gradual acceleration, coast-down, heating, cooling, pressure, and power, with sensor noise; telemetry is stored in SQLite.
- Operator HMI with START, STOP, PAUSE, and APPLY TARGET controls, live process history, command history, and a machine-condition badge (alert only, limits 80 degrees C, 5 bar, 3000 RPM).
- Four test events that create labelled ground truth for evaluation.

6.2 Part 2: network security

- Four fixture-backed adapters that normalise to one schema, with stable hash-based event identifiers.
- Network anomaly model trained on a normal baseline file, saved, and reused; separate holdout, benign, and malicious fixtures exist for testing.
- Offline replay tool with optional scoring and loopback publishing; no live capture, no PLC connection, no blocking.

6.3 Part 3: physical machine learning

- Reusable detector class with training, saved model, streaming inference, interpolation of short missing runs, and a strict rule that labels are never used as features.
- Synthetic data generators for normal telemetry and seven scenarios; separate fixed seed (2026) and different parameter ranges for the test set.
- Evaluation scripts producing accuracy, precision, recall, F1, false-positive rate, confusion matrices, per-scenario metrics, detection delay, and error listings, plus graphs.

6.4 Parts 4 and 5: correlation, risk, and OT monitor

- Event validation, duplicate suppression, asset-and-time correlation, risk calculation, and a persistent incident register that survives restarts.
- REST endpoints to submit events, read current risk, list incidents, and acknowledge them.
- Part 5 adds sensor validation (range and staleness), engineering rules (limits, rate of change, fluctuation, operating-state checks), and optional scoring with the Part 3 model.

6.5 Admin console

- Separate Flask application with hashed passwords, rate-limited login, and three roles: administrator, analyst, and operator.
- Incident review with acknowledge, escalate, false-positive marking, notes, and a simulated-response record; every decision is written to an audit log.
- Live dashboard polling every three seconds with OFFLINE and STALE indicators; a failed-login detector (five failures in five minutes) protects the console itself.

6.6 Part 6: operator-confirmed simulated response

Automated response is outside the scope of this research. Part 6 adds only an optional, human-in-the-loop simulated stop: it is offered solely for an active CRITICAL incident, needs an explicit operator confirmation with a page-generated token, and only works when both the caller and PLC address are loopback. No detector, rule, or score can trigger it.

6.7 Engineering quality

- About 8,300 lines of Python (excluding tests) and 126 automated test functions in 22 test files covering every part.
- A one-command launcher for the four services, a staged P2 to P3 to P4 to P5 chain runner with one JSON hand-off per stage, and a connected demo runner.
- Safety controls by design: loopback-only binding, ingest token, refusal of non-loopback hosts, and documentation stating that nothing here is a safety function.

7. First Experimental Results

All figures below are computed from actual model predictions on synthetic data generated for this project. They describe the physical anomaly model (Part 3) only. The fused network-plus-physical evaluation has not yet been run, so no claim about RQ4 is made here.

7.1 Overall performance of the physical model

Metric	Baseline test set (200 samples)	Harder test set (320 samples)
Accuracy	0.640	0.466
Precision	0.659	0.702
Recall	0.580	0.386
F1-score	0.617	0.499
False-positive rate	0.300	0.360
TP / TN / FP / FN	58 / 70 / 30 / 42	85 / 64 / 36 / 135
Detection delay (detected scenarios)	1 sample	1 sample (4 of 6 detected)

Table 2. Current Isolation Forest results. Delay is measured in samples, not milliseconds.

7.2 Detection by scenario type (harder test set)

Scenario	Anomalous samples	Detected	Detection rate
Sudden change	30	28	93.3%
Unstable behaviour	30	29	96.7%
Heavy sensor noise	40	26	65.0%
Gradual deviation	40	2	5.0%
Slow gradual drift	40	0	0.0%
Stealthy, in normal range	40	0	0.0%

Table 3. Per-scenario detection. Normal samples: 100, of which 36 were flagged (false positives).

7.3 Interpretation

- **Strength:** abrupt and unstable changes are detected almost always and within one sample.
- **Weakness 1:** slow drift and stealthy in-range changes are almost invisible to a model that scores each observation against the normal range. This is a known limit of point-wise anomaly detection.
- **Weakness 2:** a false-positive rate of 30 to 36 percent is too high for a useful alert stream. Likely causes are the contamination setting, the mismatch between the training and test generators, and the small test size; these will be investigated rather than assumed.

- **Why this supports the research idea:** the scenarios the physical model misses are exactly those where a network event (for example an unauthorised parameter write) may be the only early evidence. This gives a concrete, testable hypothesis for the fusion experiment.

An earlier headline figure of 97 percent accuracy and 4 percent false-positive rate came from 200 samples produced by the same generator for training and testing. It is kept in the repository only as a historical record and is **not** used as a result in this proposal.

8. Design Decisions and Scope Refinements

The table lists the places where the implementation differs from the initially planned design, with the reason for each. Reporting these openly keeps the results interpretable.

Topic	Initial plan	Current implementation	Reason or plan
Network monitoring	Live Suricata and/or Zeek on simulated traffic	Adapters and model on replayed, sanitised fixtures; no live capture	Time and resources; live capture added in Section 11
ML input features	RPM, temperature, pressure and their rates	Motor speed, temperature, vibration and their rates	Part 3 was built on a generic motor dataset; Part 1 telemetry has pressure and power but no vibration. To be aligned (Section 11)
Risk levels	LOW, MEDIUM, HIGH	LOW, MEDIUM, HIGH, CRITICAL	A fourth level separates urgent cases; the response feature depends on it
Risk formula	R = f(network, physical, criticality, correlation strength)	(0.40 N + 0.60 P) x criticality; no separate correlation term	Simplicity first; correlation term proposed and will be compared
Dashboard	Flask and Plotly	Flask with HTML, CSS, and JavaScript; no Plotly	Lighter; Plotly-style time-series views remain an option
Response	None automatic; optional recommendation	Operator-confirmed simulated stop for CRITICAL incidents	Human stays in control; no automatic action
Scenarios	S1 to S4	S1 to S4 supported; synthetic physical scenarios extended to seven	Harder cases expose model limits
Extra components	Not planned	Admin console, roles, audit log, failed-login detector	Supervisory review and protection of the platform itself

9. Evaluation Plan

The decisive experiment is the comparison of three detectors on the same scenarios and the same data partitions. This is the direct test of RQ4.

Scenario	Network	Physical	Expected interpretation
S1 Normal	Normal	Normal	No alert; used to measure false positives
S2 Network-only	Anomalous	Normal	Suspicious communication; investigate (MEDIUM expected)
S3 Physical-only	Normal	Anomalous	Process anomaly without network evidence (MEDIUM expected)
S4 Correlated	Anomalous	Anomalous	Strong combined evidence (CRITICAL expected)

Scenario	Network	Physical	Expected interpretation
S5 Stealthy (new)	Anomalous	Within normal range	Tests whether network evidence catches what the ML model misses
S6 Sensor spoof (new)	Normal	Inconsistent sensors	Tests sensor-inconsistency rules

Three detectors will be compared: **network-only**, **physical-only**, and **fused**. Metrics will be computed at the incident level using the TEST MODE ground truth:

- Precision, recall, F1-score, and false-positive rate per detector.
- Detection latency (in seconds, from the PLC write or test-event time to the alert), and lead time of the fused alert over the physical-only alert.
- Correlation rate: the share of related network and physical events that are correctly joined.
- Alert volume per hour of normal operation, as a measure of operator workload.
- Repetition over several random seeds, reported with mean and spread rather than a single run.

10. Planned Enhancements and Additional Ideas

The enhancements below follow directly from the first results. Each is small enough to be realistic, and they are listed in priority order.

10.1 Priority A: needed to answer the research questions

A1. Align the physical model with the real simulator telemetry

Retrain the Isolation Forest on the variables that Part 1 actually produces: RPM, target-versus-actual RPM deviation, temperature, pressure, and power, with their rates of change. This removes the vibration dependency, lets the live database feed be scored directly, and matches the intended design of the research.

A2. Fix the false-positive rate and use a proper data split

Separate training, validation, and test data; tune contamination and the decision threshold on validation data only; and generate the baseline from several runs of the live simulator so that it includes normal start, ramp, and stop behaviour. The target is a false-positive rate that an operator could tolerate.

A3. Add drift-aware detection for slow and stealthy changes

Add rolling-window features (mean, spread, slope) and a simple drift detector such as EWMA or CUSUM beside the Isolation Forest. Add a **physics-consistency check** that compares expected and observed behaviour, for example whether temperature and power are consistent with the current RPM. A value can be inside its normal range and still be wrong for the current operating state, which is how stealthy changes hide.

A4. Live passive Modbus monitoring on the loopback network

Replace replayed fixtures with a passive monitor of the real simulator traffic. Rules: write function codes from sources not on an allow-list, writes to the target-RPM register outside the normal range or rate, unusual write frequency, and unexpected new clients. Run Zeek or Suricata in the virtual machine if resources allow, otherwise keep a lightweight Python monitor. The research need is reliable structured evidence, not the number of tools.

A5. Full ablation experiment

Run the network-only, physical-only, and fused comparison of Section 10 end to end, driven by TEST MODE ground truth and scripted network scenarios, across several seeds.

10.2 Priority B: strengthen the contribution

B1. Add correlation strength to the risk model

The current formula does not reward correlation explicitly. Test a version with a bounded correlation bonus (for example based on time gap and event relevance) against the present formula, and report which separates S2, S3, and S4 better.

B2. Risk sensitivity analysis

Vary the weights, the thresholds, and the correlation window (for example 5, 10, and 20 seconds) and report how precision, recall, and latency change. This turns "experimental choices" into evidence about how fragile or robust they are.

B3. Explainability of every alert

For each incident show which features moved most, the rule or network indicator that fired, the time gap between events, and the arithmetic behind the score. This supports the claim that CHEM-SHIELD is transparent and not a black box.

B4. Map scenarios to MITRE ATT&CK for ICS

Describe each laboratory scenario with a matching ATT&CK for ICS technique, such as Modify Parameter (T0836), Unauthorized Command Message (T0855), and Manipulation of Control (T0831), so that the results use a recognised vocabulary. The identifiers should be verified against the current ATT&CK for ICS release when the final text is written.

10.3 Priority C: optional if time permits

- **Comparison models:** One-Class SVM, Local Outlier Factor, or a small autoencoder as optional comparison against Isolation Forest, only after the main pipeline is stable.
- **Time-series views:** Plotly or equivalent charts of telemetry, scores, and incidents in the Admin console.
- **Reproducibility package:** container or script-based setup, fixed seeds, a published synthetic dataset, and an automated test run, so that another student can repeat every experiment.
- **Analyst workload metrics:** time to acknowledge and the false-positive marking rate recorded by the Admin console.
- **Short threat model of the platform itself:** what an attacker could do to the HMI, publisher, or console, and how the existing token, loopback, rate-limit, and audit controls respond.

11. Work Plan

Phase	Activities	Status
1. Foundation	Virtual PLC, process simulator, HMI, telemetry store	Completed
2. Detection layers	Network adapters and model; Isolation Forest; OT monitor rules	Completed (first version)
3. Fusion and interface	Correlation, risk engine, incident register, dashboards, Admin console, Part 6	Completed
4. Model improvement	Items A1 to A3: feature alignment, data split, tuning, drift detection	Next
5. Live network evidence	Item A4: passive Modbus monitor and optional Zeek or Suricata	Next
6. Main experiment	Item A5 and B1 to B2: ablation, correlation term, sensitivity analysis	Planned
7. Writing and delivery	Results tables, figures, explainability views, thesis text, final demonstration	Planned

Phases 4 to 6 are ordered so that the research questions are answered first. If time becomes short, Priority C items are dropped without weakening the core result.

12. Scope, Safety, Ethics, and Limitations

12.1 Safety and dual-use controls

- All experiments occur in an isolated virtual laboratory; all services stay on 127.0.0.1 and the launcher refuses other hosts.
- No scanning, probing, or connection to any operational military, industrial, or commercial network or device.
- No malware, weaponised exploit payloads, or persistent offensive tools. Abnormal activity is limited to what is needed to produce detection evidence.
- No automatic response. The only response feature is a simulated stop that a person must confirm for a CRITICAL incident.
- Results are laboratory findings and say nothing about the security of any real facility, including any in Ethiopia.

12.2 Limitations

- The environment is simulated and synthetic telemetry may not capture real equipment variability.
- The current ML model has a high false-positive rate and misses slow or stealthy changes (Section 8).

- Network evidence is currently replayed from fixtures, so network-side performance is not yet measured live.
- Rule-based correlation on one asset and a fixed window may not generalise to multi-stage, multi-asset incidents.
- Risk weights and thresholds are design choices, not validated standards, and scores are not probabilities.
- Test sets are small and synthetic; conclusions will be stated with matching caution.

13. Final Deliverables

Deliverable	**Status**
Isolated OT simulation (virtual PLC, process model, sensors, HMI)	Delivered
Network-monitoring component producing structured events	Delivered on fixtures; live version planned
Baseline and controlled-anomaly datasets	Delivered (synthetic); live-baseline dataset planned
Isolation Forest anomaly detection module	Delivered; improvement planned
Rule-based correlation engine	Delivered
Documented risk-scoring engine	Delivered; sensitivity study planned
Monitoring dashboard and Admin console	Delivered
Unified event and data model	Delivered
Experimental results and evaluation tables	Physical model done; fused comparison remaining
Source code, configuration, datasets, screenshots, diagrams, thesis	Code delivered; screenshots and thesis text remaining

14. Conclusion

CHEM-SHIELD rests on one focused idea: a valid-looking network event and an unusual physical state are more meaningful together than apart. The project already has a working prototype that covers the full chain from a simulated machine to an operator warning, with automated tests and safety controls. The first results show honestly that the physical model needs improvement for slow, stealthy, and noisy cases, and that the false-positive rate must come down. These findings do not weaken the research; they define it. The planned enhancements target the weak points, and the planned fused comparison will provide a measured answer to the central question.

Appendix A: Repository Layout

Folder	Contents
chem_shield_p1/	Virtual PLC, process simulator, operator HMI, telemetry store
chem_shield_p2/	Network adapters, network anomaly model, replay tool, fixtures
chem_shield_p3/	Physical Isolation Forest, data generators, evaluation, results, graphs
chem_shield_p4/	Correlation engine, risk engine, incident store
chem_shield_p5/	OT physical monitor: sensor validation, rules, model scoring
chem_shield_p6/	Warning popup and operator-confirmed simulated response
chem_shield_common/	Shared loopback publisher with ingest token
admin_part/	Admin console, user management, audit log, login detector
scripts/	One-command launcher, staged chain runner, demo runner

Appendix B: End-to-End Demonstration Procedure

The final demonstration will follow this sequence:

Step	Action	Expected result
1	Start the PLC, process simulator, HMI, and Admin console (python scripts/run_all.py)	All services online on loopback
2	Press START and run at a normal target RPM	Stable telemetry; no incident
3	Inject a controlled network event (for example an out-of-range target write)	Network event with a score
4	Observe the process change or use a TEST MODE event	Physical anomaly with a score
5	Correlation links both events (same asset, within 10 s)	One correlated incident
6	Risk engine scores the incident	Higher level than either event alone
7	HMI shows the warning; Admin console shows evidence, scores, and level	Operator alert with explanation
8	Analyst acknowledges; operator may confirm a simulated stop if CRITICAL	Decision recorded in the audit log; no automatic action